

UNIT - 1

Page No.

Date: 15/01/18

Electromagnetic Field Theory (EMFT)

1) Gradient

3) Curl

2) Divergence

1) GRADIENT: $\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$
(del)

$\phi(x, y, z) = \text{Scalar field}$

$$\nabla \phi(x, y, z) = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \phi(x, y, z)$$

$$\Rightarrow \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z} \equiv \text{gradient of scalar function}$$

Gradient of scalar function is defined as the maximum rate of change of scalar quantity along the direction of vector field.

OR

It is defined as the gradient of any scalar function as a vector whose direction and magnitude are those of the maximum rate of change of ϕ .

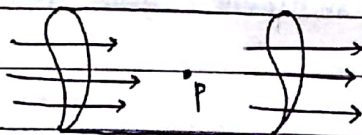
2) DIVERGENCE: $\vec{A} = \text{vector field}$

$$\vec{A} = \hat{i} A_x + \hat{j} A_y + \hat{k} A_z$$

$$\nabla \cdot \vec{A} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (\hat{i} A_x + \hat{j} A_y + \hat{k} A_z)$$

$$\equiv \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \equiv \text{Scalar field}$$

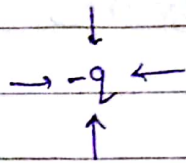
Divergence is the term which is related to liquids.



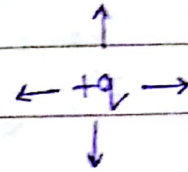
divergence at P = $\text{div. } \vec{P} = 0$

For incompressible liquid the incoming water through a tube is equal to the outgoing water.

- In case of -ve charge divergence is minimum.



- In case of +ve charge divergence is maximum.



- Divergence means dot product of 2 vector mathematically. But according to the Physics, it means spreading of electric-field intensity w.r.t a +ve charge.
- Divergence: It is the limiting value of net outward flow of some physical quantity like liquid or electric flux through the surface area of a unit volume.
- If Divergence of any vector is zero then the vector is called solenoidal vector.

3) CURL: Rotation of any vector quantity.

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \Rightarrow \hat{i} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) - \hat{j} \left(\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z} \right) + \hat{k} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

$\vec{A} \rightarrow$ any vector quantity

$$+ \hat{k} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

Curl of a vector field is a vector whose magnitude is max., net circulation of magnetic field per unit area as the area tends to zero.

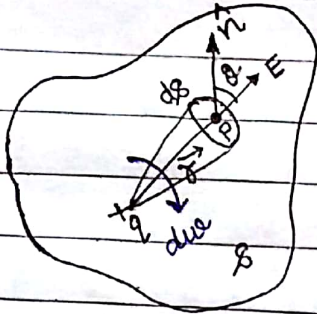
Curl $\leftarrow (\nabla \times \vec{A}) = 0$ then $\vec{A} \rightarrow$ irrotational vector

- 1) $\nabla \cdot (\nabla \phi) =$ Divergent of gradient of a scalar function.
- 2) $\nabla \times (\nabla \times \vec{A}) =$ Curl of curl of a vector quantity.
- 3) $\nabla(\nabla \cdot \vec{A}) =$ Gradient of divergence of vector function.

19/01/18

Gauss - Law in integral form

Electric flux: $\Phi_E = \oint_S \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$



$$d\Phi_E = \vec{E} \cdot d\vec{s} \quad \text{--- (1)}$$

$$\oint_S d\Phi_E = \oint_S \vec{E} \cdot d\vec{s}$$

$$= \oint_S \frac{q}{4\pi\epsilon_0 r^2} \cdot \hat{r} \cdot d\vec{s} = \oint_S \frac{q}{4\pi\epsilon_0 r^2} (\hat{r} \cdot \hat{n}) d\Omega$$

$$\oint_S d\Phi_E = \oint \frac{q}{4\pi\epsilon_0 r^2} \cdot \cos\theta \cdot d\Omega$$

- $d\Omega$ is the solid angle i.e. the projection of 2 quantities.

$$= \frac{q}{4\pi\epsilon_0} \oint \frac{d\Omega \cdot \cos\theta}{r^2}$$

solid angle in 3D

$$\Phi_E = \frac{q}{4\pi\epsilon_0} \oint d\Omega = \frac{q}{4\pi\epsilon_0} \times 4\pi = \frac{q}{\epsilon_0}$$

- Gauss divergence theorem Volume integral into surface integral

$$\iiint_V (\nabla \cdot \vec{A}) \cdot dV = \iint_S \vec{A} \cdot d\vec{s}$$

Volume Integral

Surface Integral

- Stoke's Theorem: Surface integral into line integral.

$$\iint_S (\nabla \times \vec{A}) \cdot d\vec{s} = \oint_L \vec{A} \cdot d\vec{l}$$

line Integral

or

$$\iint_S (\text{curl } \vec{A}) \cdot d\vec{s} = \oint \vec{A} \cdot d\vec{l}$$

$\rho \rightarrow$ linear charge density

$\sigma \rightarrow$ surface charge density

$\rho \rightarrow$ Volume charge density.

$$\bullet \lambda = \frac{dq}{dl}$$

$$\Rightarrow dq = dl \cdot d\lambda$$

$$\bullet \sigma = \frac{q}{S} \Rightarrow q = \sigma \cdot S$$

$$\bullet \rho = \frac{q}{dV}$$

$$\Rightarrow q = \rho \cdot dV$$

Gauss law in Differential form

$$\Phi_E = \oint_S \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$$

By Generalising point charge in 3D: $q = \rho \cdot dV$

$$\Phi_E = \oint_S \vec{E} \cdot d\vec{s} = \iiint_V \frac{\rho \cdot dV}{\epsilon_0} \quad - (1)$$

Applying Gauss divergence theorem to electric field:

$$\oint_S \vec{E} \cdot d\vec{s} = \oint_V (\nabla \cdot \vec{E}) \cdot dV \quad - (2)$$

eq. (1) and eq. (2):

$$\oint_V (\nabla \cdot \vec{E}) \cdot dV = \iiint_V \frac{\rho \cdot dV}{\epsilon_0} \quad - (3) \Rightarrow \oint_V \frac{\rho \cdot dV}{\epsilon_0}$$

$$\oint_V \left[(\nabla \cdot \vec{E}) - \frac{\rho}{\epsilon_0} \right] dV = 0 \quad - (4)$$

For arbitrary volume this integral becomes zero.

Gauss law in Differential form

$$\boxed{\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}} \quad - (5)$$

Divergence of electric field = $\frac{1}{\epsilon_0}$ times total volume charge density over a closed volume

Continuity Equation

$$\Phi_E = \oint_S \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$$

$$\begin{aligned} \Phi_E = \oint_S \vec{E} \cdot d\vec{s} &= \oint_S E \cdot \hat{s} \cdot d\vec{s} \\ &= E \oint_S \hat{s} \cdot \hat{n} \cdot d\vec{s} \end{aligned}$$

$$\Rightarrow \oint \cos \theta \cdot ds = E \cdot 4\pi r^2 \cos \theta$$

$$\vec{F} = q\vec{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{qQ_1}{r^2} \cdot \hat{r}$$

Coulombs law
using test charge.

20/10/18

- Q1) Find the gradient of a scalar function of $V(x, y, z) = x^2y + e^z$.
Calculate the magnitude of gradient at point $(1, 5, -2)$.
- Q2) Find the value of constant c for which vector $\vec{A} = (x+3y)\hat{a}_x + (y-2z)\hat{a}_y + (x+cz)\hat{a}_z$ is solenoidal.
- Q3) Find the gradient of $r = \sqrt{x^2 + y^2 + z^2}$
- Q4) If $\phi(x, y, z) = 3(x^2y - y^2x)$. Calculate gradient at $(1, -2, -1)$
- Q5) Given a vector $\vec{A} = \hat{i}(x^2 - y^2) + \hat{j}(x + y) + \hat{k}(x^2 - z^2)$. Calculate curl at $\vec{A}$ at $(\frac{1}{2}, 3, 1)$.

Sol.ⁿ 1) $V(x, y, z) = x^2y + e^z$

$$\nabla V = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x^2y + e^z)$$

$$\left[\frac{1}{e^2} = 0.135 \right]$$

$$\nabla V = 2xy\hat{i} + x^2\hat{j} + e^z\hat{k}$$

$$(x, y, z) \rightarrow (1, 5, -2)$$

$$\text{Magnitude of gradient} = \sqrt{4x^2y^2 + x^4 + e^{2z}}$$

$$= \sqrt{100 + 1 + \frac{1}{e^4}} = \sqrt{101 + \frac{1}{e^4}}$$

$$= 11.135$$

Sol.ⁿ 2) $\nabla \cdot \vec{A} = 0$

$$\left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left((x+3y)\hat{a}_x + (y-2z)\hat{a}_y + (x+cz)\hat{a}_z \right) = 0$$

$$\hat{i} \cdot \hat{a}_x + \hat{j} \cdot \hat{a}_y + c(\hat{k} \cdot \hat{a}_z) = 0$$

$$c = -2$$

Sol.ⁿ 3) $r(x, y, z) = \sqrt{x^2 + y^2 + z^2}$

$$\nabla r = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left(\sqrt{x^2 + y^2 + z^2} \right)$$

$$\Rightarrow \hat{i} \cdot \frac{1}{2\sqrt{x^2+y^2+z^2}} \cdot 2x + \hat{j} \cdot \frac{1}{2\sqrt{x^2+y^2+z^2}} \cdot 2y + \hat{k} \cdot \frac{1}{2\sqrt{x^2+y^2+z^2}} \cdot 2z$$

$$\Rightarrow \frac{1}{\sqrt{x^2+y^2+z^2}} (x\hat{i} + y\hat{j} + z\hat{k}) = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{|r|} = \frac{\vec{r}}{|r|} = \hat{r}$$

Q4) $\nabla \cdot \phi = \hat{i} \cdot 3(2xy - y^2) + \hat{j} [3(x^2 - 2xy)] + \hat{k} \cdot 3(0)$
 $= 3\hat{i}(2xy - y^2) + 3(x^2 - 2xy)\hat{j}$
 putting $(x, y, z) \rightarrow (1, -2, -1)$
 $= 3(-4 - 4)\hat{i} + 3(1 + 4)\hat{j}$
 $= -24\hat{i} + 15\hat{j}$

$$\text{Magnitude} = \sqrt{576 + 225} = 28.3$$

Sol. 5) $\vec{A} = \hat{i}(x^2 - y^2) + \hat{j}(x + y) + \hat{k}(x^2 - z^2)$

$$\nabla \times \vec{A} = \hat{i} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) - \hat{j} \left(\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z} \right) + \hat{k} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

$$A_x = (x^2 - y^2) \quad A_y = (x + y) \quad A_z = x^2 - z^2$$

$$\Rightarrow \hat{i} [0 - 0] - \hat{j} [2x - 0] + \hat{k} [1 + 2y]$$

$$\Rightarrow -2x\hat{j} + 2y\hat{k} + \hat{k}$$

$$\Rightarrow -\hat{j} + 6\hat{k} + \hat{k} = -\hat{j} + 7\hat{k}$$

Q. Curl (grad ϕ) = 0. Show that.

Sol. 7) $\nabla \times \nabla \phi = 0$

$$\text{L.H.S} = \text{R.H.S}$$

$$\left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times \left(\hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z} \right) = 0 \quad \text{: To prove}$$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial z} \right) - \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial y} \right) \right] - \hat{j} \left[\frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial z} \right) - \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial x} \right) \right] + \hat{k} \left[\frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial y} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial x} \right) \right] = 0$$

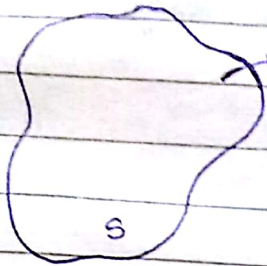
*** Continuity Equation

It is based on law of conservation of charge.

It is the relation b/w. current density and charge density.

$\vec{J}$ = current density

ρ = volume charge density.



Volume; V

Total current inside this closed volume = $\oint_S \vec{J} \cdot d\vec{s}$ (I) ①

considering a wire then current passing through it

$$\Rightarrow I = -\frac{dq}{dt} = -\frac{d}{dt} \oint_V \rho \cdot dV \quad \text{--- (2)}$$

$\rho \rightarrow$ volume charge density

The rate of increase of total charge inside any arbitrary volume must be equal to net flow of charge in that volume.

An equation which expresses equality of incoming and outgoing charges in a system is called continuity equation.

$\frac{dq}{dt} = I$ i.e. current in wire is due to electric charge carriers

from eq. ① and ②:

$$\oint_S \vec{J} \cdot d\vec{s} = -\frac{d}{dt} \oint_V \rho \cdot dV \quad \text{--- (3)}$$

Applying Gauss Divergence Theorem to eq. ①:

(Applicable only to vector quantities)

$$\oint_S \vec{J} \cdot d\vec{s} = \oint_V (\nabla \cdot \vec{J}) dV \quad \text{--- (4)}$$

$$\oint_V (\nabla \cdot \vec{J}) dV = -\oint_V \left(\rho \frac{\partial}{\partial t} \right) dV \quad \text{--- (5)}$$

$$\oint_V \left(\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} \right) dV = 0 \quad \rightarrow$$

For an arbitrary volume, this integral is zero.

$$\boxed{\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0}$$

$\rightarrow$ continuity Equation

for a steady state, i inside a closed loop is zero,
 $i = 0$

$$\nabla \cdot (\text{current density}) = 0$$

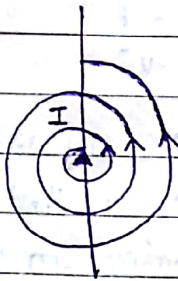
$$\Rightarrow \nabla \cdot \vec{J} = 0$$

current density over a closed loop is zero.
 $\Rightarrow$ law of conservation of charge.

Maxwell Equation

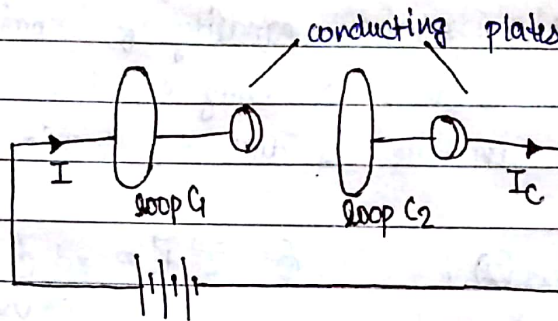
Gauss law in Magnetic field:

Ampere's law: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$

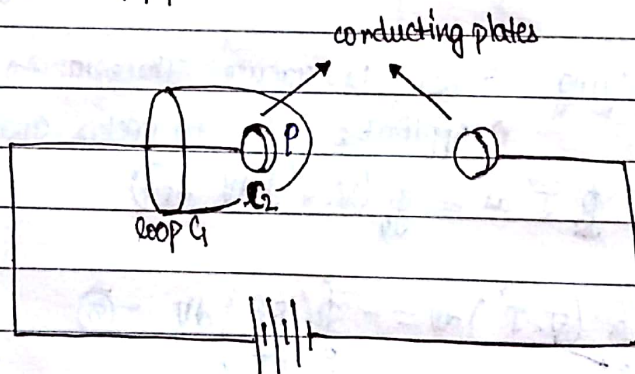


Maxwell Modified Displacement Law:

If we apply Ampere's law in 11^{th} plate capacitor then it becomes inconsistent.



Magnetic field is produced in loop C_1 due to varying current which produces the electric field.



Time varying magnetic field produces electric field.

$$\oint_{\text{loop } C_1} \vec{B} \cdot d\vec{l} = \mu_0 I \quad \text{--- (1)}$$

$$\oint_{\text{loop } C_2} \vec{B} \cdot d\vec{l} = 0 \quad \text{--- (2)} \quad \text{As loop } C_2 \text{ is not connected to the circuit.}$$

current passing through circuit is conduction current.
 Total current inside the circuit $\rightarrow I = I_c + I_d$

I_d is due to time varying magnetic field which gives electric flux.

$$I_d = \epsilon_0 \cdot \frac{d\phi_E}{dt}$$

Eq. (1) and (2) should continue to hold if loop C_1 and loop C_2 are closed to plate P .

$$\therefore \oint_{\text{loop } C_1} \vec{B} \cdot d\vec{l} = \oint_{\text{loop } C_2} \vec{B} \cdot d\vec{l} \quad - (3)$$

eq. (3) is contradictory to eq. (1) and eq. (2) that's why we have to modify amperes law.

Displacement current: I_d is that current which comes into existence due to change in electric flux with time.

Modified form of Ampere's law	$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left[I_c + \epsilon_0 \frac{d\phi_E}{dt} \right]$
-------------------------------	---

29/01/18

Derivation of Maxwell's Equation

Gauss Law of electricity:

1) $\nabla \cdot \vec{D} = \rho$ - (1)

2) $\nabla \cdot \vec{B} = 0$ - (2) : Gauss's law of magnetism

Faraday's law of electromagnetic induction:

3) $\nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$ - (3)

Ampere's law for currents in conductors

4) $\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$ - (4)

$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$

$(\vec{D} = \epsilon_0 \vec{E})$

Electric Displacement vector

(1) $\nabla \cdot \vec{D} = \rho$

Derivation of eq. (1): $\oint \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0} = \phi_E$ - (1)

for a boundary surface S

having volume $V \Rightarrow \oint \vec{E} \cdot d\vec{s} = \frac{1}{\epsilon_0} \int \rho \cdot dV$ - (2)

In the dielectric Medium,

$\rho = \rho + \rho_{\text{Polarisation}}$ Total charge density is the combination of polarised charge carriers

and free charge carriers.

$$\oint \vec{E} \cdot d\vec{s} = \frac{1}{\epsilon_0} \int (\rho + \rho_{pol.}) dV \quad (3)$$

electric field is opposite to the bound charge carriers.

$$\text{div } \vec{P} = \rho_{pol.}$$

where $\vec{P}$ is the polarisation vector.

$$\oint \vec{E} \cdot d\vec{s} = \frac{1}{\epsilon_0} \int (\rho - \text{div} \cdot \vec{P}) dV$$

$$\oint_s (\epsilon \cdot \vec{E}) \cdot d\vec{s} = \int \rho dV - \int (\text{div } \vec{P}) dV \quad (4)$$

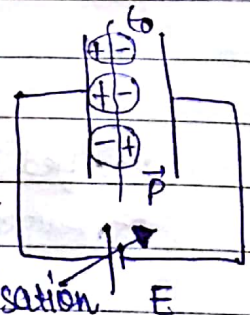
Applying Gauss Divergence theorem;

$$\oint \text{div} (\epsilon \vec{E}) dV = \int \rho dV - \int (\text{div } \vec{P}) dV$$

$$\oint \text{div} (\epsilon \vec{E} + \vec{P}) dV = \int \rho dV$$

$$\oint \text{div } \vec{D} dV = \int \rho dV$$

For bound charge carriers: $\epsilon \vec{E} + \vec{P} = \vec{D}$



* Polarisation in dielectric is directly proportional to the applied voltage is the one which leads to distribution of charge inside dielectric.

$$\oint_V [\text{div } \vec{D} - \rho] dV = 0$$

For arbitrary volume, this integral vanishes.

$$\text{div} \cdot \vec{D} - \rho = 0$$

$$\boxed{\nabla \cdot \vec{D} = \rho}$$

$\vec{D}$ related to external magnetic field. Higher the electric field higher will be $\vec{D}$. For free charge carriers,

polarisation 'is zero. $\epsilon_0 \vec{E} + 0 = \vec{D}$

$$\boxed{\vec{D} = \epsilon_0 \vec{E}}$$

Divergence is scattering or losing free charge carriers.

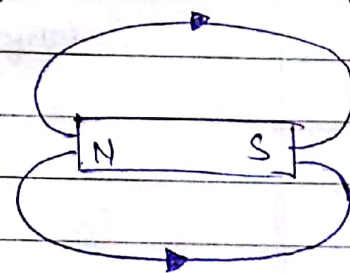
$\vec{B} \rightarrow$ Magnetic Flux Density.

(2) $\nabla \cdot \vec{B} = 0$ (# In Dot Product, Apply Gauss Divergence Theorem)

Magnetic field over a closed loop is always zero as the no. of magnetic lines entering any arbitrary volume are equal to the no. leaving it.

It means monopoles never exist.

$$\int \vec{B} \cdot d\vec{s} = 0 \quad \text{--- (1)}$$



Applying Gauss Divergence Theorem:

$$\int (\text{div. } \vec{B}) dV = 0$$

For arbitrary volume, this integral becomes zero.

$$\text{div. } \vec{B} = 0$$

$$\boxed{\nabla \cdot \vec{B} = 0}$$

Conclusion: Monopoles never exist. Hence proved.

$$(3) \quad \nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

This eq.ⁿ is based on Faraday's law of Electromagnetic Induction. In cross product or curl, we will apply Stoke's Theorem.

* The rate of change of magnetic flux in a circ. produces an induced emf in that circuit.

$$e = - \frac{d\Phi_B}{dt} \quad \text{--- (1)}$$

• Time varying magnetic field produces the electric field.

magnetic flux, $\phi_B = \oint_S \vec{B} \cdot d\vec{s}$

putting eq. (1): $\epsilon = -\frac{d}{dt}(\oint \vec{B} \cdot d\vec{s})$ — (2)

$$\epsilon = \oint_l \vec{E} \cdot d\vec{l} = -\frac{d}{dt}(\oint \vec{B} \cdot d\vec{s})$$

where dl is small elemental length.

* Induced emf is equal to the line-integral of Electric field.

Applying Stoke's Theorem to convert line integral into surface integral:

$$\oint (\nabla \times \vec{E}) \cdot d\vec{s} = -\oint \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$$

$$\oint (\nabla \times \vec{E} + \frac{\partial \vec{B}}{\partial t}) \cdot d\vec{s} = 0$$

For arbitrary surface, this integral vanishes.

So, $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

Hence Proved.

02/02/18 4) $\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$

(Related to ampere's Law)

$$\oint_l \vec{B} \cdot d\vec{l} = -\text{Total change passing through the conductor.}$$

Ampere's Law: $\oint \vec{H} \cdot d\vec{l} = I = \oint \vec{J} \cdot d\vec{s}$ — (1)

(sum of conduction + Displacement current) $\hookrightarrow$ current density

$$\vec{D} = \epsilon \vec{E}$$

$$\vec{B} = \mu_0 \vec{H}$$

Apply Stoke's Theorem to convert line integral into surface integral.

$$\oint (\nabla \times \vec{H}) \cdot d\vec{s} = \oint \vec{J} \cdot d\vec{s}$$

Ampere's law is applicable only to steady state current. Not for varying electric field. Thus, it is inconsistent.

$$\oint (\nabla \times \vec{H}) - \vec{J} \cdot d\vec{s} = 0$$

$$\nabla \times \vec{H} = \vec{J} \quad \text{--- (1)}$$

eq. (1) is inconsistent as it is applicable for steady current not applicable to time varying electric field.

$$\nabla \times \vec{H} = \vec{J} + \vec{J}' \quad \text{--- (2)}$$

eq. (2) is applicable for only time-varying electric field.

Taking Divergence on both sides of eq. (2):

$$\text{div. (curl } \vec{H}) = \nabla \cdot \vec{J} + \nabla \cdot \vec{J}'$$

($\therefore$ Div. (curl of vector quantity) is always zero.)

$$0 = \nabla \cdot \vec{J} + \nabla \cdot \vec{J}'$$

$$\nabla \cdot \vec{J}' = -\nabla \cdot \vec{J}$$

using continuity eq. $\Rightarrow \nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$

$$\frac{\partial \rho}{\partial t} = \nabla \cdot \vec{J}'$$

$$\frac{\partial (\nabla \cdot \vec{D})}{\partial t} = \nabla \cdot \vec{J}'$$

$$\Rightarrow \nabla \cdot \vec{J}' = \nabla \cdot \left(\frac{\partial \vec{D}}{\partial t} \right)$$

$$\vec{J}' = \frac{\partial \vec{D}}{\partial t}$$

putting the value of $\vec{J}'$ in eq. (2),

$$\Rightarrow \boxed{\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}}$$

Modified Form of Maxwell Eq. in differential form.

• Derivation in integral form:

1) Differential Form

3) Stoke's law

2) Gauss law

4) Integral form.

(7 marks)

Poynting Theorem

$$\oint \text{div}(\vec{E} \times \vec{H}) \cdot dV = -\frac{\partial}{\partial t} \int \left[\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right] \cdot dV - \int \sigma E^2 \cdot dV$$

in a medium not in free space

Total power leaving
the volumeRate of decrease in
energy stored in electric
and Magnetic fields.Ohmic
losses.
(Total ohmic
power
dissipation
within the system)

- It is based on law of conservation of electromagnetic field.
- If we consider the total value of electromagnetic field then it is related to the total power in the closed volume.

Derivation:

$$\nabla \cdot \vec{D} = \rho \quad \text{--- (1)}$$

$$\nabla \cdot \vec{B} = 0 \quad \text{--- (2)}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{--- (3)}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \quad \text{--- (4)}$$

Taking scalar product of magnetic field
w.r.t (3) and electric field w.r.t (4).

$$\vec{H} \cdot (\text{curl } \vec{E}) = -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} \quad \text{--- (5)}$$

$$\vec{E} \cdot (\text{curl } \vec{H}) = \vec{E} \cdot \vec{J} + \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \quad \text{--- (6)}$$

Using identity $\text{div}(\vec{E} \times \vec{H}) = \vec{H} \cdot \text{curl } \vec{E} - \vec{E} \cdot \text{curl } \vec{H}$

This identity is related to scalar-Triple product, (scalar-vector product).

$$\nabla \cdot (\vec{E} \times \vec{H}) = \text{div}(\vec{E} \times \vec{H})$$

Subtract eq. (5) and eq. (6):

$$\vec{H} \cdot (\text{curl } \vec{E}) - \vec{E} \cdot (\text{curl } \vec{H}) = -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} - \vec{E} \cdot \vec{J} - \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \quad \text{--- (7)}$$

$$\text{div}(\vec{E} \times \vec{H}) = -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} - \vec{E} \cdot \vec{J} - \vec{E} \cdot \frac{\partial \vec{D}}{\partial t}$$

$$\vec{B} = \mu \vec{H} \quad \text{and} \quad \vec{D} = \epsilon \vec{E}$$

$$\text{div.}(\vec{E} \times \vec{H}) = -\mu \cdot \mu \frac{\partial H}{\partial t} - \vec{E} \cdot \vec{J} - \vec{E} \cdot \epsilon \frac{\partial \vec{E}}{\partial t}$$

partially differentiating $H \cdot \frac{\delta H}{\delta t} \Rightarrow \frac{1}{2} \frac{\delta H^2}{\delta t}$
w.r.t time of magnetic field

as it is in
a medium not
free space

$$\text{div.}(\vec{E} \times \vec{H}) = -\mu \cdot \frac{1}{2} \frac{\delta H^2}{\delta t} - \vec{E} \cdot \vec{J} - \frac{1}{2} \epsilon \frac{\delta E^2}{\delta t} \quad \text{--- (8)}$$

taking volume integral on both sides of this eq. (8)

$$\oint \text{div.}(\vec{E} \times \vec{H}) \cdot dV = -\frac{\delta}{\delta t} \oint \left(\frac{1}{2} \mu H^2 + \frac{1}{2} \epsilon E^2 \right) dV - \oint (\vec{E} \cdot \vec{J}) dV$$

we know $\vec{J} = \sigma \vec{E}$

(Ohm's law in Electromagnetic field)

(For a wire $\Rightarrow V = IR$)

$$\Rightarrow \oint \text{div.}(\vec{E} \times \vec{H}) \cdot dV = -\frac{\delta}{\delta t} \oint \left(\frac{1}{2} \mu H^2 + \frac{1}{2} \epsilon E^2 \right) dV - \oint \sigma E^2 \cdot dV$$

voltage is related to $\vec{E}$

$$\vec{J} = \frac{I}{A}$$

$\sigma \rightarrow$ conductivity

3/02/18

Wave Equation for plane electromagnetic wave in vacuum

Q. Show that EM wave in free space travel with speed of light.

let us consider a medium characterised by ϵ_0 and μ_0 .

If charge is zero then current, $I=0$ thus $\vec{J}=0$ and $\rho=0$.

Maxwell's Equations: $\nabla \cdot \vec{D} = \rho$ --- (1) $\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$ --- (4)

$$\nabla \cdot \vec{B} = 0 \quad \text{--- (2)}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{--- (3)}$$

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} \quad \text{--- (5)}$$

Apply free-space condition to these equations:

$$\nabla \cdot \vec{D} = 0 \quad \text{--- (6)}$$

$$\nabla \cdot \vec{B} = 0 \quad \text{--- (7)}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{--- (8)}$$

taking curl on both sides of eq. (3)

$$\Rightarrow \nabla \times \nabla \times \vec{E} = -\frac{\delta}{\delta t} (\nabla \times \vec{B}) = -\mu_0 \cdot \frac{\delta}{\delta t} (\nabla \times \vec{H})$$

(as $\vec{B} = \mu_0 \vec{H}$ and $\vec{D} = \epsilon_0 \vec{E}$)

$$\Rightarrow -\mu_0 \frac{\delta}{\delta t} \left(\frac{\delta \vec{D}}{\delta t} \right) = -\mu_0 \epsilon_0 \frac{\delta^2 \vec{E}}{\delta t^2}$$

[Using $\vec{A} \times \vec{B} \times \vec{C} = \text{vector Triple Product} = \vec{B} \cdot (\vec{A} \cdot \vec{C}) - \vec{C} \cdot (\vec{A} \cdot \vec{B})$]

$$\nabla \times \nabla \times \vec{E} = \nabla (\underbrace{\nabla \cdot \vec{E}}_0) - \vec{E} (\nabla^2) = -\vec{E} (\nabla^2)$$

$$\times \vec{E} (\nabla^2) = \times \mu_0 \epsilon_0 \frac{\delta^2 \vec{E}}{\delta t^2}$$

⑨

(* Eq. n wot electric field.)

Similarly,

$$\nabla^2 (\vec{H}) = \mu_0 \epsilon_0 \frac{\delta^2 \vec{H}}{\delta t^2}$$

-⑩

eq. ⑨ and eq. ⑩ are the wave equations in free space.

General wave equation :

$$\nabla^2 \psi = \frac{1}{v^2} \frac{\delta^2 \psi}{\delta t^2}$$

-⑪

$$\frac{1}{v^2} = \mu_0 \epsilon_0$$

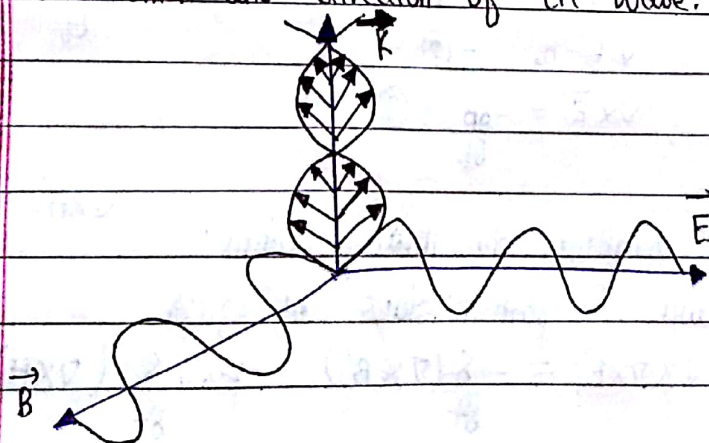
$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

where $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$$

$$v = 2.99 \times 10^8 \text{ m/s}$$

Q. Prove that em field vectors $\vec{E}$ and $\vec{H}$ are mutually $\perp^r$ to each other and direction of EM wave.



$\vec{k}$ is the wave propagation vector and represent the output vector.

$$\vec{k} = \hat{n} k = \hat{n} \cdot 2\pi$$

→ Difference b/w. 2 conjugative waves.

$$\vec{k} = \hat{n} \frac{2\pi\gamma}{c} = \hat{n} \cdot \frac{\omega}{c}$$

$$\vec{k} = \frac{\omega}{c} \cdot \hat{n}$$

Solution of wave Equations are:

$$\left. \begin{aligned} E &= E_0 \cdot e^{i(\vec{k} \cdot \vec{r} - \omega t)} \\ H &= H_0 \cdot e^{i(\vec{k} \cdot \vec{r} - \omega t)} \end{aligned} \right\} \text{--- (2)}$$

E_0 and H_0 maximum amplitude of wave:-

$$\nabla \cdot \vec{E} = 0 \quad \text{and} \quad \nabla \cdot \vec{H} = 0$$

$$\left(\hat{i} \cdot \frac{\partial}{\partial x} + \hat{j} \cdot \frac{\partial}{\partial y} + \hat{k} \cdot \frac{\partial}{\partial z} \right) \cdot \left[E_0 \cdot e^{i(\vec{k} \cdot \vec{r} - \omega t)} \right]$$

$$\vec{k} \cdot \vec{r} = (\hat{i} k_x + \hat{j} k_y + \hat{k} k_z) \cdot (x\hat{i} + y\hat{j} + z\hat{k}) = xk_x + yk_y + zk_z$$

$$\nabla \cdot \vec{E} \Rightarrow i(\vec{k} \cdot \vec{E}) \Rightarrow \vec{k} \cdot \vec{E} = 0$$

Electric field is $\perp^r$ to wave propagation vector $\vec{k}$, $\vec{k} \cdot \vec{E} = 0$
Similarly, $\vec{k} \cdot \vec{H} = 0$

3/2/18

Q. Transverse Nature of EM w.r.t $\vec{k}$.

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\mu_0 \frac{\partial \vec{H}}{\partial t} \quad [\text{w.r.t (3)rd Maxwell eq.}]$$

$$= -\mu_0 \cdot \frac{\partial}{\partial t} \left[H_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \right]$$

$$= -\mu_0 [H_0 \cdot e^{i(\vec{k} \cdot \vec{r} - \omega t)} \cdot (-i\omega)]$$

$$\nabla \times \vec{E} = i\omega H_0 \vec{H}$$

$$\nabla \cdot \vec{E} = i\vec{k} \cdot \vec{E}$$

$$\text{Similarly, } \nabla \times \vec{E} = i\vec{k} \times \vec{E}$$

$$\Rightarrow i(\vec{k} \times \vec{E}) = i\omega H_0 \vec{H}$$

$$\boxed{\vec{k} \times \vec{E} = \mu_0 \omega \vec{H}} \text{--- (3)}$$

conclusion: magnetic field is mutually $\perp^r$ to $\vec{k}$ and $\vec{E}$

Similarly, $\boxed{\vec{K} \times \vec{H} = -\epsilon_0 \omega \cdot \vec{E}}$ (w.r.t eq. ④ of Maxwell's Equation)

($\therefore$ replace $\vec{E}$ by $\vec{H}$)

Eq. ③ and eq. ④ represent:

Transverse Nature of EM w.r.t K.

Q. Prove that $\vec{E}$ and $\vec{H}$ are in same phase.

From eq. (13): $\vec{H} = \frac{1}{\mu_0 \omega} (\vec{K} \times \vec{E}) = \frac{K}{\mu_0 \omega} (\hat{n} \times \vec{E})$

$$= \frac{1}{\mu_0 c} \cdot E$$

$$\frac{E}{H} = \mu_0 c = \frac{E}{H}$$

$$\left| \frac{E}{H} \right| = \mu_0 c = \frac{\mu_0 \cdot 1}{\sqrt{\mu_0 \epsilon_0}} = \sqrt{\frac{\mu_0}{\epsilon_0}} \rightarrow \text{real and constant}$$

$$\left| \frac{V/m}{A/m} \right| = \sqrt{\frac{\mu_0}{\epsilon_0}} = \left| \frac{V}{A} \right| \begin{matrix} \text{Voltage} \\ \text{Current} \end{matrix} \Rightarrow \text{Resistance}$$

$\hookrightarrow$ units $\rightarrow$ electric field component

$$\left| \frac{E}{H} \right| = \left| \frac{E_0}{H_0} \right| = Z_0 \text{ (Impedance in a EM wave)}$$

$$= \sqrt{\frac{4\pi \times 10^{-7} \text{ H/m}}{8.85 \times 10^{-12} \text{ Farad/m}}}$$

Only in case of free space: $\left| \frac{E}{H} \right| = 376.6 \Omega = Z_0$

As $\sqrt{\frac{\mu_0}{\epsilon_0}} \Rightarrow$ real and constant. Thus component of E & H is in same phase.

Energy flow due to plane EM in free space:

$$\text{Poynting Vector} = \vec{S} = \vec{E} \times \vec{H}$$

$$\vec{S} = \vec{E} \times \left[\frac{1}{\mu_0 \omega} (\vec{k} \times \vec{E}) \right]$$

from eq. (B) and (A) $\Rightarrow \vec{H} = \frac{\hat{n} \times \vec{E}}{\mu_0 c}$

$$\vec{S} = \vec{E} \times \left(\frac{\hat{n} \times \vec{E}}{\mu_0 c} \right)$$

$$= \frac{1}{\mu_0 c} (\vec{E} \times \hat{n} \times \vec{E}) \rightarrow \text{vector triple product}$$

$$= \frac{1}{\mu_0 c} [\hat{n} E^2] = \left(\frac{1}{Z_0} \right) [\hat{n} E^2]$$

$$= \frac{E^2}{Z_0} \cdot \hat{n}$$

Avg. value of Poynting Vector over a complete cycle $\Rightarrow \langle S \rangle$

$$\langle S \rangle = \frac{1}{Z_0} \langle E^2 \rangle \hat{n}$$

Total energy flow per unit area per unit time:

$$\Rightarrow \frac{1}{Z_0} \langle E_0^2 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \rangle_{\text{real}} \cdot \hat{n}$$

$$\Rightarrow \frac{1}{Z_0} \cdot E_0^2 \langle \cos^2(\vec{k} \cdot \vec{r} - \omega t) \rangle \cdot \hat{n}$$

$$\boxed{\text{as } e^{i\theta} = \cos\theta + i\sin\theta}$$

($\therefore$ Avg. value of $\cos^2 = \frac{1}{2}$)
over a complete cycle.

$$\langle S \rangle = \frac{1}{Z_0} \cdot \frac{E_0^2}{2} \cdot \hat{n}$$

$$\langle S \rangle = \frac{E^2}{Z_0} \cdot \text{r.m.s.} \cdot \hat{n}$$

($\therefore$ Energy flow per unit A per unit time lies along the dir.ⁿ of propagation of wave)

Ratio of electric field energy density to magnetic field energy density.

$$\frac{U_e}{U_m} = \frac{\frac{1}{2} \epsilon_0 E^2}{\frac{1}{2} \mu_0 H^2} = \frac{\epsilon_0}{\mu_0} \cdot \left| \frac{E}{H} \right|^2 = \frac{\epsilon_0}{\mu_0} \cdot \left| \sqrt{\frac{\mu_0}{\epsilon_0}} \right|^2 = \frac{\epsilon_0}{\mu_0} \cdot \frac{\mu_0}{\epsilon_0} = 1$$

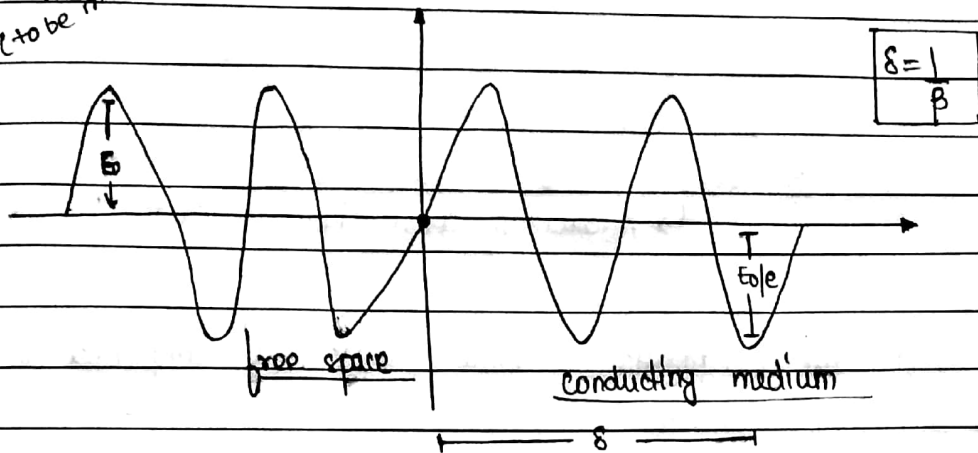
It means electric & magnetic field are equally shared in the propagation of wave.

$$\begin{array}{l} \text{Electro-static} \\ \text{Energy stored} \end{array} = \begin{array}{l} \text{Magnetic-static} \\ \text{Energy stored} \end{array}$$

$$\boxed{U_e = U_m}$$

Skin Depth: Propagation of EM wave in conducting medium $\rightarrow$
 During the propagation the losses are present in em wave,
 (attenuation)

**
 (to be made in exam)



δ = skin depth or Penetration depth.

Properties inside the conducting medium: $\epsilon, \mu, \vec{J} = \sigma \vec{E}, \rho = 0$.

In free space, $\rho = 0$ and $\vec{J} = 0$.

Charge inside
 conductor is zero.

Maxwell's Eq.ⁿs in conducting Medium:

$$\nabla \cdot \vec{D} = 0$$

$$\nabla \times \nabla \times \vec{E} = -\frac{\partial (\nabla \times \vec{B})}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\frac{\partial (\nabla \times \mu_0 \vec{H})}{\partial t}$$

$$\nabla \times \vec{H} = \sigma \vec{E} + \frac{\partial \vec{D}}{\partial t}$$

(from 1st Maxwell eq.ⁿ)

$$-\nabla^2 \vec{E} = -\mu \cdot \frac{\partial (\nabla \times \vec{H})}{\partial t}$$

putting the value of $\nabla \times \vec{H}$ from 4th Maxwell eq.ⁿ

$$-\nabla^2 \vec{E} = -\mu \frac{\partial}{\partial t} \left[\sigma \vec{E} + \frac{\partial \vec{D}}{\partial t} \right]$$

$$-\nabla^2 \vec{E} = -\mu \cdot \sigma \cdot \frac{\partial \vec{E}}{\partial t} - \mu \epsilon \cdot \frac{\partial^2 \vec{E}}{\partial t^2}$$

Similarly,

$$\nabla^2 \vec{H} = \mu \sigma \cdot \frac{\partial \vec{H}}{\partial t} + \mu \epsilon \cdot \frac{\partial^2 \vec{H}}{\partial t^2}$$

Date _____

$$E = E_0 [e^{i(\vec{k} \cdot \vec{r} - \omega t)}]$$

$$H = H_0 [e^{i(\vec{k} \cdot \vec{r} - \omega t)}]$$

$\vec{k} = \omega/c$ (in free space), $\vec{k} = \omega/v$ (in non-conducting dielectric medium)

$\vec{k} \rightarrow$ complex quantity in case of conducting medium.

$$k = \alpha + i\beta$$

$\hookrightarrow$ Attenuation const. i.e. related to skin depth.

Find $\alpha = ?$ and $\beta = ?$

- During the propagation of emw the total amplitude is reduced to $\frac{E_0}{e}$ (initial amplitude).

In case of free space, $\alpha = \beta = 0$.

- The depth at which electromagnetic waves entering a conducting medium, is damped to $1/e$ of its initial amplitude at its surface.

$$\delta = \frac{1}{\beta} = \frac{1}{\omega \sqrt{\mu \epsilon}} \left[\frac{\sqrt{1 + (\sigma/\omega \epsilon)^2} - 1}{2} \right]^{-1/2}$$

For conductors, $\frac{\sigma}{\omega \epsilon} \gg 1 \Rightarrow \sigma \gg \omega \epsilon$

$$\delta = \sqrt{\frac{2}{\omega \sigma \mu}}$$

where $\omega \rightarrow$ angular frequency $= 2\pi\nu$

$\sigma \rightarrow$ conductivity of the medium.

$$\alpha = \frac{1}{\omega \sqrt{\mu \epsilon}} \left[\frac{\sqrt{1 + (\sigma/\omega \epsilon)^2} + 1}{2} \right]^{-1/2}$$

Q. EM wave is travelling in z-dir.ⁿ in a dielectric medium with $\mu_r = 1$ and $\epsilon_r = 2$. It has peak electric field strength $E_0 = 5 \text{ V/m}$. Find

- ① Impedance in medium
- ② Peak magnetic field strength
- ③ velocity of wave.

Sol.ⁿ ① $Z = \frac{|E_0|}{|H_0|} = \sqrt{\frac{\mu}{\epsilon}}$ (in medium) $= \sqrt{\frac{\mu_r \cdot \mu_0}{\epsilon_r \cdot \epsilon_0}} = \sqrt{\frac{\mu_r}{\epsilon_r}} \cdot \sqrt{\frac{\mu_0}{\epsilon_0}}$

$$\mu_r = \frac{\mu}{\mu_0}$$

$$Z \Rightarrow 376.6 \times \sqrt{\frac{1}{2}}$$

Relative value
of permeability

$$\text{Impedance, } Z = 266.8 \, \Omega$$

② $Z = \frac{|E_0|}{|H_0|} \Rightarrow 266.8 = \frac{5}{|H_0|}$

$$\downarrow E_0 = 5 \text{ V/m}$$

max. electric field

$$H_0 = 1.87 \times 10^{-2} \text{ A/m}$$

③ $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$ (in free space)
(speed of light) $\sim 3 \times 10^8 \text{ m/s}$

In dielectric medium, $c = \frac{1}{\sqrt{\mu \epsilon}}$

putting $\mu = \mu_r \cdot \mu_0 \Rightarrow c = \frac{1}{\sqrt{\mu_r \mu_0 \cdot \epsilon_r \epsilon_0}}$

$$c = \frac{1}{\sqrt{\mu_r \mu_0}} \times 3 \times 10^8$$

$$\text{speed of wave} = 2.12 \times 10^8 \text{ m/s}$$

UNIT-2

12/02/18

Quantum Mechanics

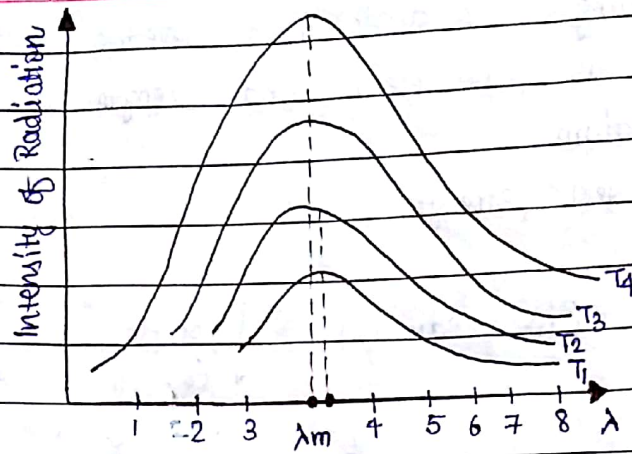
Planck's Radiation Law:

Black body Radiations $\rightarrow a+b+c=1$

$$a=b=c$$

$$a=1$$

Date _____



$$T_1 = 2000\text{K}$$

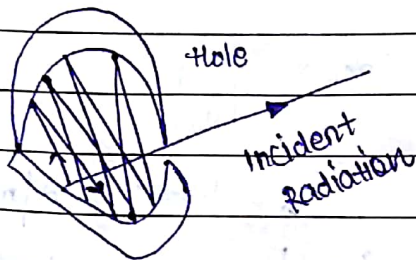
$$T_2 = 3000\text{K}$$

$$T_3 = 4000\text{K}$$

$$T_4 = 5000\text{K}$$

$\lambda_m, T = \text{constant}$
Weir's Law

→ Applicable for shorter wavelength.



$$[\text{Thermal Energy} = K_B \cdot T]$$

$K_B \rightarrow$ Boltzmann constant

$T \rightarrow$ Room Temperature.

$$E \propto T^4 \Rightarrow \boxed{E = \sigma T^4}$$

Derivation: let N be total no. of oscillators and E is their energy :

$$N = N_0 + N_1 + N_2 + \dots + N_x$$

N_0 = no. of oscillators in ground state,

N_1 = no. of oscillators in 1st state and so on.

$$\Rightarrow E = 0 + EN_1 + 2EN_2 + \dots + xEN_x$$

Acc. to maxwell distribution law:

$$N_x = N_0 e^{-xE/K_B T}$$

$$N = N_0 + N_0 e^{-E/K_B T} + N_0 e^{-2E/K_B T} + \dots$$

$$N = N_0 \left[1 + e^{-E/K_B T} + e^{-2E/K_B T} + \dots \right]$$

$$N = \frac{N_0}{1 - e^{-E/K_B T}} \left\{ \text{as } N = N_0 \left[1 + x + x^2 + \dots \right] \Rightarrow \frac{N_0}{1 - x} \right\}$$

$$E = 0 + EN_0 e^{-E/k_B T} + 2EN_0 e^{-2E/k_B T} + 3EN_0 e^{-3E/k_B T} + \dots$$

$$\Rightarrow EN_0 e^{-E/k_B T} [1 + 2e^{-E/k_B T} + 3e^{-2E/k_B T} + \dots]$$

obtained by using: $1 + 2x + 3x^2 + 4x^3 + \dots = \frac{1}{1-x^2}$

$$\therefore \text{we get } E = EN_0 e^{-E/k_B T} \cdot \frac{1}{(1 - e^{-E/k_B T})^2}$$

Avg. energy of oscillator $\Rightarrow \bar{E} = \langle E \rangle = \frac{E}{N}$

$$\Rightarrow \frac{EN_0 \cdot e^{-E/k_B T}}{(1 - e^{-E/k_B T})^2} \cdot \frac{(1 - e^{-E/k_B T})}{N_0}$$

$$\Rightarrow \frac{E \cdot e^{-E/k_B T}}{1 - e^{-E/k_B T}} = \frac{E}{e^{E/k_B T} - 1}$$

$$\left[\langle E \rangle = \frac{h\nu}{e^{h\nu/k_B T} - 1} \right]$$

No. of oscillators in frequency range ν and $\nu + d\nu$ per unit volume,

$$N = N(\nu) \cdot d\nu = \frac{8\pi\nu^2 d\nu}{c^3}$$

Total energy per unit volume in range $d\nu$:

$$E_\nu \cdot d\nu = \left[\frac{8\pi\nu^2}{c^3} \cdot \frac{h\nu}{e^{h\nu/k_B T} - 1} \right]$$